PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Effects of mHealth on the psychosocial health of pregnant women and mothers: A systematic review
AUTHORS	Sakamoto, Jennifer Lisa; Carandang, Rogie Royce; Kharel, Madhu; Shibanuma, Akira; Yarotskaya, Ekaterina; Basargina, Milana; Jimba, Masamine

VERSION 1 – REVIEW

REVIEWER	Kabongo, Eveline Stellenbosch University, Division of Health Systems and Public Health,
REVIEW RETURNED	13-Oct-2021

GENERAL COMMENTS	This is a well written paper that explained the effect of the mHealth program on pregnant women and mothers' psychosocial health. However, the paper can be improved by reorganizing some sections as following: In the first paragraph you are talking about psychosocial health during pregnancy and post-partum. In the second paragraph you should talk about the potential of mHealth in improving maternal health. The background offers no critical evaluation of mHealth, which is presented as almost entirely as though there are no barriers or potential problems with its use. The third paragraph should focus on the potential barriers to mHealth program in Maternal health. (use single studies) The paragraph four should focus only on the systematic review studies on mHealth, do not mixt single studies and systematics reviews in the same paragraph. Many of the references are old, please provide the up to date references. The abstract show that you used narrative synthesis, but in the methodology section this is not well stated. The study design, should be described in order to adequately describe the process used to undertake the narrative synthesis. Please make it clear for your reader by explaining what you mean by narrative synthesis and all it steps. Synthesis Without Meta-analysis (SWiM) reporting items should be referred to as an additional file or table in the paper. You are talking about S3 Table, but this table is situated at the bottom of the paper. If you are refereing to the SWiM please make it clear. Some tables number are missing in the additional file. All tables used should be referred to in the paper. If the table is on the additional file this should be referred to as (e.g see Table 3 Additional file 1)

	More attention should be given to tables, on page 16 line 2 -16 there is a table with no heading and this table is not reported in the paper. The discussion section is away too long, please try to shorten it by focusing the main findings only. Thank you

REVIEWER	Değirmenci, Filiz Mersin University, Nursing
REVIEW RETURNED	14-Oct-2021

GENERAL COMMENTS	I think that the article will make useful contributions to the literature on the subject. May be published as is.

REVIEWER	Patel, Suha Brigham and Women's Hospital, Division of Global Obstetrics and Gynecology
REVIEW RETURNED	07-Nov-2021

GENERAL COMMENTS	Consider changing the term "pregnant women" to "pregnant people" and include "pregnant person" in participant search. If not able to change this, then please state in limitations that nonbinary terms were not included. You briefly mention need for further research on psychosocial health of same sex and transgender partners in conclusions.

REVIEWER	Wallwiener, Stephanie University of Heidelberg
REVIEW RETURNED	08-Nov-2021

GENERAL COMMENTS	General comments: Peripartum mental disorders are highly prevalent and impose a major burden for the expectant mother, her environment, and the health care systems. Pregnant women engage regularly with digital health technology and are willing to participate in perinatal interventions delivered electronically via SMS or smartphone apps. mHealth interventions may represent a promising and effective approach to alleviate psychosocial stress. This paper gives an overview about the effects of mHealth interventions on the psychosocial health or pregnant women and mothers. The process and the results of the systematic literature search performed are clearly presented. It is well-written, and the importance of the topic is clear. The main limitation is the heterogeneity and variability of the studies included, so that a narrative synthesis was conducted instead of a meta-analysis. Major comments: This paper reviewed the evidence on the effectiveness of mHealth on psychosocial health. The main outcomes should be refined in more detail. What does a “positive effect” mean, e.g., relief of depressive/anxiety symptoms, increase in life satisfaction/quality of life? The authors also included changes in high-risk health behaviors (improvement in nutritional intake and physical activity); this should be clearly stated. The results section lacks an overview of the different types of mHealth interventions. Table S7 provides an overview of the characteristic of the included studies and interventions. However, only SMS and app services are distinguished. What types of

	interventions were used, e.g., simple text reminders via SMS, self-monitoring or home-measurement systems, virtual visits, social interactive components, cognitive behavioral therapy (CBT) based or psycho-educational approaches? The heterogeneity of the interventions presented is high, which should be expressed more clearly. In the discussion section, the authors mention that mHealth interventions could be particularly useful for a disadvantaged population. Indeed, women from disadvantaged groups often have poor treatment adherence to traditional care and high attrition rates. Possible reasons should be provided to support the potential of mHealth interventions in special populations. In the limitations section, the authors should elaborate on the heterogeneity of the included studies. There is a wide range of study designs, sample sizes (n=5 up to n=1422) and measures. Outcome measures were quite varied, e.g., depression was assessed with the EPDS, Hospital Anxiety and Depression Scale (HADS), Center for Epidemiologic Studies Depression Scale (CES-D), and Beck Depression Inventory II, making comparison of results difficult. A recent review by Chan et al in JMIR (2019) showed that positive effects were often mentioned in qualitative studies, and interventions with smaller sample sizes tended to report larger effect sizes. This should be kept in mind when interpreting the results. Minor comments: There are a few minor errors or sentences that are hard to understand, consider proof-reading. - Table 1: Last column should be renamed, e.g. “total number” instead of “total times” outcome reported - Page 10, lines 252-255: consider restructuring of sentence. - Page 11, line 262: one word missing - Page 11, line 292, one word missing, also consider restructuring of sentence. - Page 13, line 333, rephrase sentence, “postpartum women with gestational DM”, sounds confusing

VERSION 1 – AUTHOR RESPONSE

Response to Reviewer #1:
1) In the first paragraph you are talking about psychosocial health during pregnancy and post-partum. In the second paragraph you should talk about the potential of mHealth in improving maternal health. The background offers no critical evaluation of mHealth, which presented as almost entirely as though there are no barriers or potential problems with its use. The third paragraph should focus on the potential barriers to mHealth program in maternal health (use single studies). The paragraph four should focus only on the systematic review studies on mHealth, do not mix single studies and systematic reviews in the same paragraph.

Thank you very much for the helpful and constructive advice. We have organized and revised the background section following your suggestion.

In the second paragraph, we have added more information to discuss the potential of mHealth interventions in improving maternal health. (Page 4, Lines 93-95)

“mHealth apps can also support pregnant women and mothers to manage their own health, promote a healthy lifestyle, and encourage access to information at any time and place.[15]”

In the third paragraph, we only included single studies to discuss the potential barriers of mHealth interventions among pregnant women and mothers. (Page 4, Lines 96-104)

“Despite the potentials of mHealth, there are also its challenges that remain to be tackled. In low-resource settings, potential barriers to mHealth interventions include the limited level of literacy, access to mobile data, knowledge of technology, cultural beliefs, and availability of mobile devices.[16] …… Other barriers include unstable power supply and poor infrastructure and connectivity to internet, especially in rural or conflict-affected areas.[17, 18]”

In the fourth paragraph, we exclusively discussed one systematic review study by Dol et al., 2020. (Page 4, Lines 105-118)

“A study by Dol et al. systematically reviewed the impact of mHealth interventions during the perinatal period on maternal psychosocial health outcomes.[20]….”

2) Many of the references are old, please provide the up to date references.

We appreciate your kind advice. We understand that including new references is important to provide up to date research results. Therefore, we have cited references where most (aside from studies included in the analysis) are published later than 2015. For those that were published prior to 2015 are studies that we believed to be most appropriate and up to date in providing necessary insight to the paper after conducting literature review.

3) The abstract show that you used narrative synthesis, but in the methodology section this is not well stated. The study design should be described in order to adequately describe the process used to undertake the narrative synthesis. Please make it clear for your reader by explaining what you mean by narrative synthesis and all its steps.

Thank you very much for the comment. We have added more information on the data synthesis. (Page 7, Lines 190-202)

“A detailed examination was conducted on the numeric and textual summary findings of the included studies. Findings were then synthesized narratively and the studies were grouped according to psychosocial health outcomes. Summary of findings was presented in a table including psychosocial health outcome, types of mHealth services, total number the outcome was reported, and whether the findings had no effect, mixed effect, or positive effect. Conclusions were reached in each study for the effects of mHealth intervention. ….”

4) Synthesis Without Meta-analysis (SWiM) reporting items should be referred to as an additional file or table in the paper. You are talking about S3 Table, but this table is situated at the bottom of the paper. If you are referring to the SWiM please make it clear.

We have included the SWiM guidelines as Supplementary File 4 and cited in the main text. (Page 7, Lines 189-190)

“Instead, we utilized Synthesis Without Meta-analysis (SWiM) reporting guidelines (see Supplementary File 4) to conduct data synthesis.[37]”

5) Some table numbers are missing in the additional file. All tables used should be referred to in the paper. If the table is on the additional file, this should be referred to as (e.g., see Table 3 Additional file 1).

Thank you very much for the comment. To avoid any confusion of the supplementary files, we have revised the names of the supplementary files and made sure to cite all of them within the main text (e.g., see Supplementary File 1).

6) More attention should be given to tables, on page 16 line 2-16 there is a table with no heading and this table is not reported in the paper.

Thank you very much for the comment. As there are no tables located on Page 16 Lines 2-16, we assume the reviewer has kindly pointed out the only table included in our main text labeled “Table 1. Summary of the effect of mHealth interventions on psychosocial health outcomes of pregnant women and mothers” which is located on Page 10. We refer to this table in the main text on Page 9 Line 265. We made sure that the table was not cut off in the middle.

7) The discussion section is away too long, please try to shorten it by focusing the main findings only.

Thank you very much for your advice. We have revised the discussion section to include only the main findings. Specifically, we have omitted paragraphs 6 and 8, combined paragraphs 2 and 3, and reduced its length while also incorporating suggestions from Reviewer #4.

Response to Reviewer #2:
1) I think the article will make useful contributions to the literature on the subject. May be published as is.

Thank you very much for your kind and encouraging comment.

Response to Reviewer #3:
1) Considering changing the term “pregnant women” to “pregnant people” and include “pregnant person” in participant search. If not able to change this, then please state in limitations that nonbinary terms were not included.

Thank you very much for your valuable insight. We have added to the study limitations that nonbinary terms were not included as search terms and therefore may have excluded nonbinary from our study. (Page 17, Lines 491-492)
“Fourth, nonbinary terms were not included as search terms and this may have excluded nonbinary people from our study.”

Response to Reviewer #4:
Major comments:
1) This paper reviewed the evidence on the effectiveness of mHealth on psychosocial health. The main outcomes should be refined in more detail. What does a “positive” effect mean, e.g., relief of depressive/anxiety symptoms, increase in life satisfaction/quality of life? The authors also included changes in high-risk health behaviors (improvement in nutritional intake and physical activity); this should be clearly-stated.

Thank you very much for pointing this out. We have added more information on how we have reached conclusion on the outcomes. Moreover, we have revised to only include psychosocial health outcomes as changes of health behavior outcomes were not the scope of the current study. (Page 7, Lines 195-202)

“We considered an outcome to have a “positive effect” if the mHealth intervention showed a significant effect (e.g., improvement in anxiety/depressive symptoms, increase in mental well-being/self-efficacy) and narrative findings indicated positive results (i.e., benefits of using mHealth services). …”

2) The results section lacks an overview of the different types of mHealth interventions. Table S7 provides an overview of the characteristic of the included studies and interventions. However, only SMS and app services are distinguished. What types of interventions were used, e.g., simple text reminders via SMS, self-monitoring or home-measurement systems, virtual visits, social interactive components, cognitive behavioral therapy (CBT) based or psycho-educational approaches? The heterogeneity of the interventions presented is high, which should be expressed more clearly.

Thank you very much for your valuable comment. We have assessed the types of mHealth interventions used in the included studies based on the mHealth and ICT Framework and a study by Chen et al 2018. We then categorized the interventions into ten types: 1) Health education or promotion, 2) Communication and support, 3) Psychoeducation or therapy, 4) Self-monitoring system, 5) Reminders, 6) Decision guideline, 7) Behavior change, 8) Laboratory results, 9) Registries/vital events tracking, and 10) Electronic health records. We have added this information in the main text (Page 9, Lines 256-262) and in the “Supplementary File 8. Characteristics of included studies".

References used:
• Labrique AB, Vasudevan L, Kochi E, et al. mHealth innovations as health system strengthening tools: 12 common applications and a visual framework. Glob Health Sci Pract. 2013;1(2): 160–171. doi:10.9745/GHSP-D-13-00031
• Chen H, Chai Y, Dong L, et al. Effectiveness and appropriateness of mHealth interventions for maternal and child health: systematic review. JMIR Mhealth Uhealth. 2018;6(1): e7. doi: 10.2196/mhealth.8998

3) In the discussion section, the authors mention that mHealth interventions could be particularly useful for a disadvantaged population. Indeed, women from disadvantaged groups often have poor treatment adherence to traditional care and high attrition rates. Possible reasons should be provided to support the potential of mHealth interventions in special populations.

We have added possible reasons for how mHealth interventions can be particularly useful in special populations. (Page 15, Lines 417-419; Page 16, Lines 447-452)

“mHealth can provide ease of access to health education materials and self-monitoring systems where the users can track their own behavior or health data while also being supported by healthcare providers.”

“Findings suggest that mHealth interventions can provide support to pregnant women and mothers who may feel hesitant to seek support in-person due to the fear of being judged. The flexible nature of mHealth interventions enables accommodating users’ specific needs and tailoring the program to their preferences (e.g., cultural, literacy, language preferences). This makes it particularly useful for vulnerable populations who often have specific physical and mental health needs.[86]”

4) In the limitations section, the authors should elaborate on the heterogeneity of the included studies. There is a wide range of study designs, sample sizes (n=5 up to n=1422) and measures. Outcome measures were quite varied, e.g., depression was assessed with the EPDS, Hospital Anxiety and Depression Scale (HADS), Center for Epidemiologic Studies Depression Scale (CES-D), and Beck Depression Inventory II, making comparison of results difficult. A recent review by Chan et al. in JMIR

(2019) showed that positive effects were often mentioned in qualitative studies, and interventions with smaller sample sizes tended to report larger effect sizes. This should be kept in mind when interpreting the results.

Thank you very much for this suggestion. We have elaborated more on the heterogeneity of the included studies and also referred to Chan et al. 2019. (Page 16, Lines 456-461)

“Studies had a wide range of study designs, sample sizes (n = 4 to n = 2,782), and outcome measures (e.g., anxiety was assessed with Generalized Anxiety Disorder-7 (GAD-7), Beck Anxiety Inventory (BAI), Hospital Anxiety and Depression Scale (HADS), which made the comparison of results difficult. Moreover, positive effects may have been emphasized in qualitative studies and studies with small sample sizes may have reported larger effect sizes.[14] Thus, the findings should be interpreted cautiously.”

Minor comments:
There are a few minor errors or sentences that are hard to understand, consider proof-reading.

Thank you very much for your thorough check and comments. We have revised the errors and sentences accordingly.

- Table 1: Last column should be renamed, e.g., “total number” instead of “total times” outcome reported.
We have renamed the last column as “total number” (Page 10).

- Page 10, lines 252-255: consider restructuring of sentence
We have restructured the sentence as “Similarly, Baumel et al. conducted a quasi-experimental study to examine the effectiveness of an app that provided self-help tools and emotional support delivered by trained volunteers to pregnant women diagnosed with postpartum depression.” (Page 11, Lines 292- 294)

- Page 11, lines 262: one word missing
We have added the missing word “information” in the sentence (Page 11, Line 302)

- Page 11, line 292: one word missing, also consider restructuring of sentence
We have restructured the sentence as “Jallo et al. conducted a quasi-experimental study to investigate the effectiveness of a stress coping app to reduce stress in a sample of pregnant women staying in an obstetrical antepartum high-risk unit.” (Page 12, Line 321-322)

- Page 13, line 333: rephrase sentence, “postpartum women with gestational DM,” sounds confusing
We have rephrased the sentence as “In Norway, women with gestational diabetes mellitus who used an app described an increase in feeling of control to manage their own health:…” (Page 13, Line 363-364)

VERSION 2 – REVIEW

REVIEWER	Kabongo, Eveline Stellenbosch University, Division of Health Systems and Public Health,
REVIEW RETURNED	31-Dec-2021

GENERAL COMMENTS	The manuscript has improved tremendously, I think it is ready for publication as I do not have more comment to add.